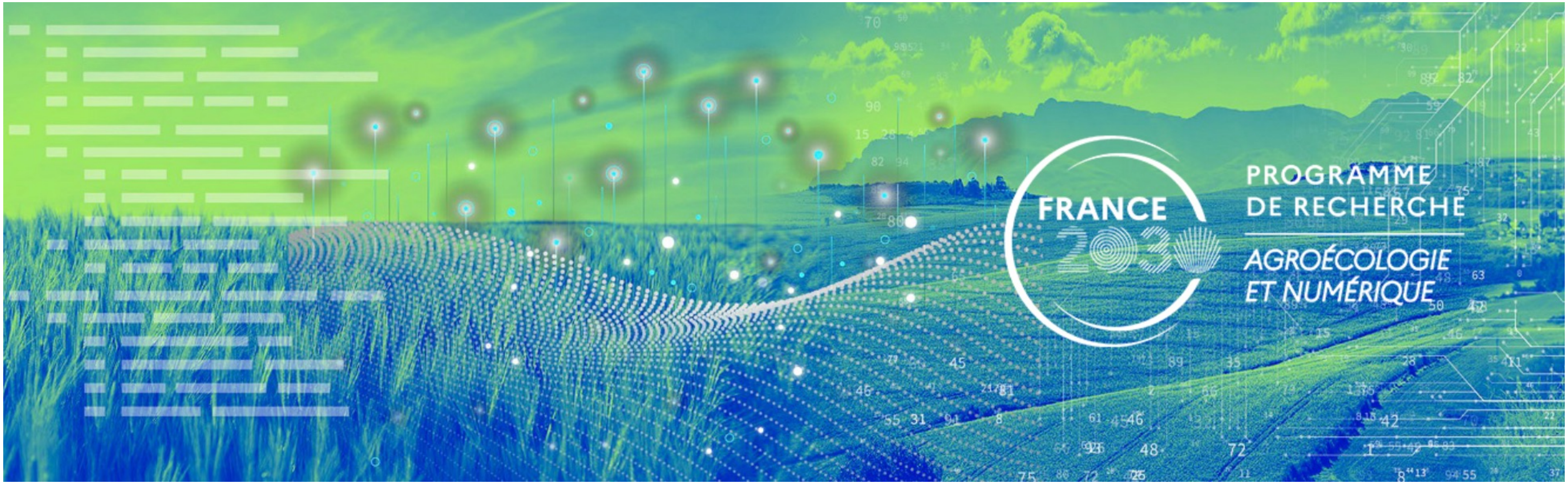

Building the LoRa IoT platform
Part 2: the customized IoT gateway

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Wireless Sensors Made Simple
for agroecology & sustainable agriculture

Powered by technologies developed in Intel-IrriS

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The customized IoT gateway

- The customized IoT gateway is based on the generic WaziGate
- WaziGate is an IoT LoRa Gateway developed by WAZIUP
- WaziGate implements the edge-enabled IoT gateway approach
  - customized applications can be directly hosted in the gateway
  - the gateway can easily work without Internet connectivity
  - data are available to end-users in an embedded database
  - web-based visualization module provides graphical user interface
- You can find all the WaziGate documentation on the WaziGate documentation page.
- The customized IoT gateway adds many more features
  - Simple configuration for all supported sensor types
  - Sensor dashboard based on Home Assistant
  - Automatic backup and simple restore procedure for sensor data
  - Display on OLED screen, …

Install software for your gateway

- **PEPR AgriFutur gateway distribution (RPI3B/3B+/4B)**
  - comes pre-configured
  - will work out-of-the box with the default sensor devices
  - Including the LoRaCAM-AI device
  - **Download the gateway SD card image from project website**
  - http://intel-irris.eu/results
  - Image uses EU868 frequency band
  - EU433 or other bands can be configured afterwards
  - Flash the SD card image on 8GB SD card class 10

Look at the video produced for INYTEL-IRRIS for the main steps
Video at t=124s: https://youtu.be/j-1Nk0tv0xM?t=124

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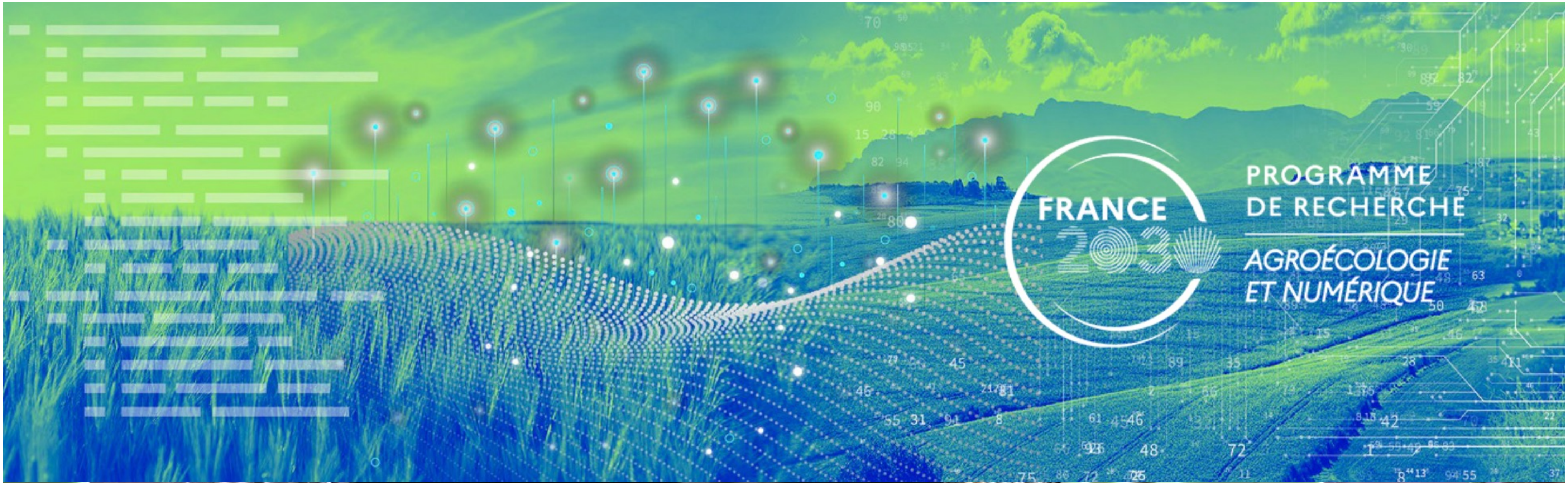

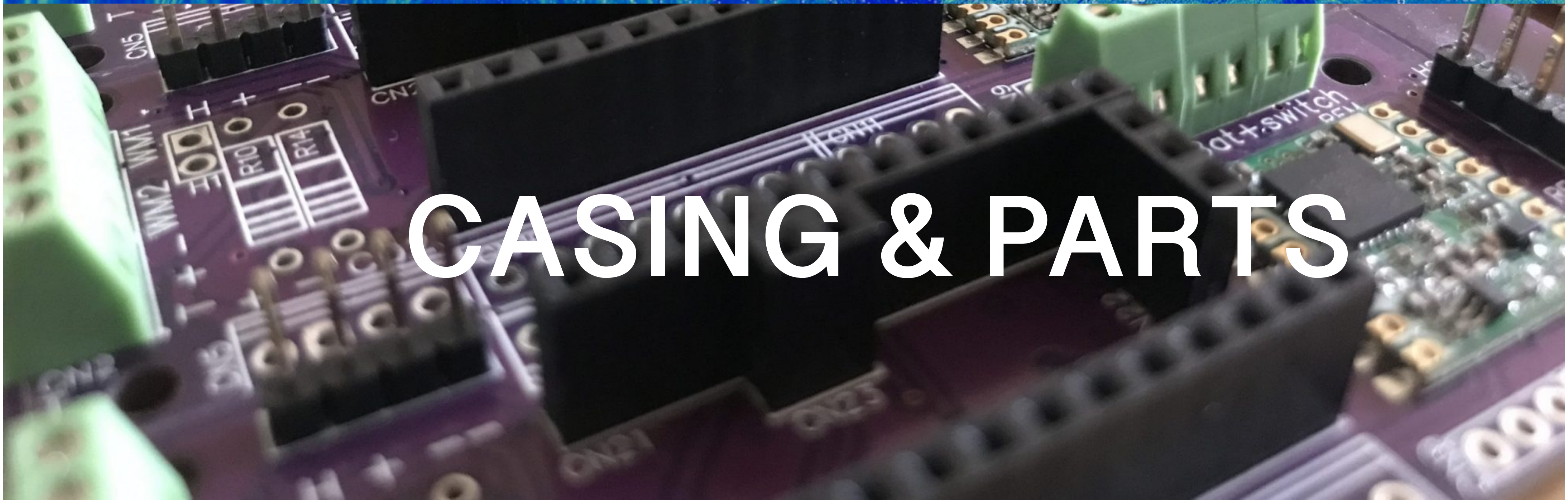

CASING & PARTS

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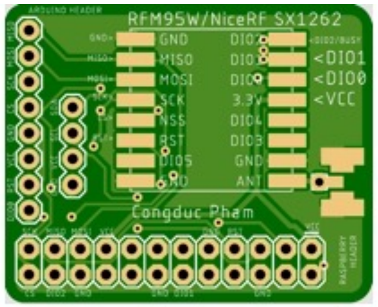
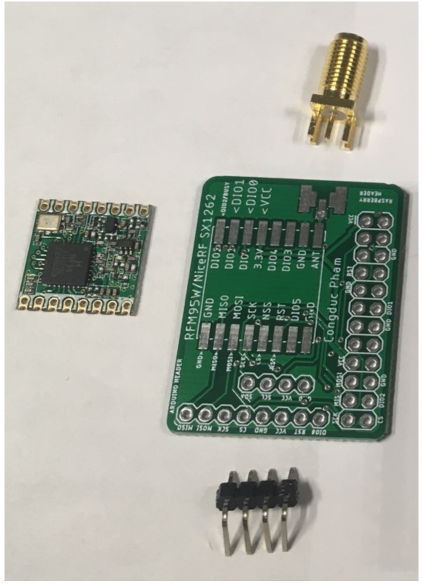
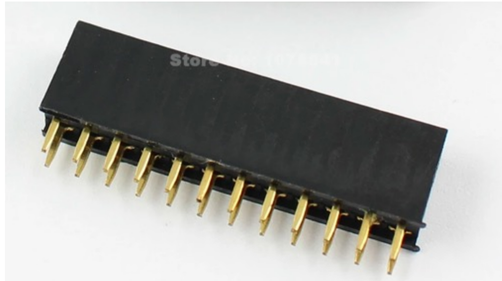
Single-channel: the LoRa radio hat

https://github.com/CongducPham/PRIMA-Intel-IrriS/blob/main/Tutorials/Intel-Irris-PCB.pdf



RFM95W/NiceRF SX1262 breakout





2-row 12-pin header
or
2 X 1-row 12-pin header

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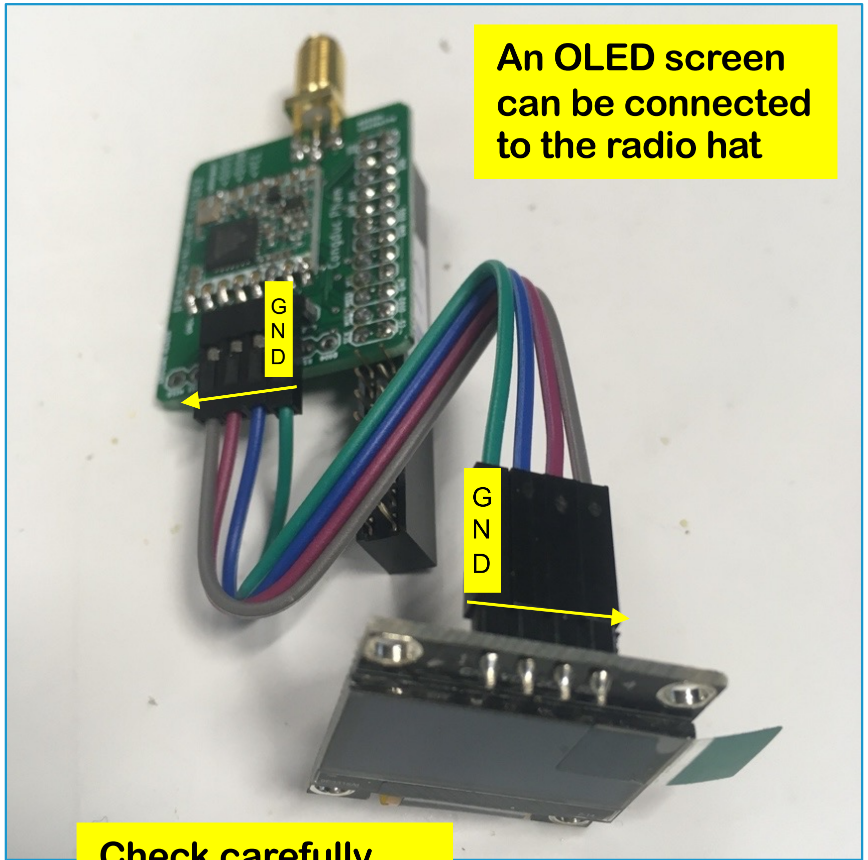
Installing OLED



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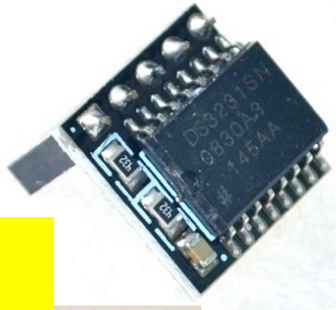
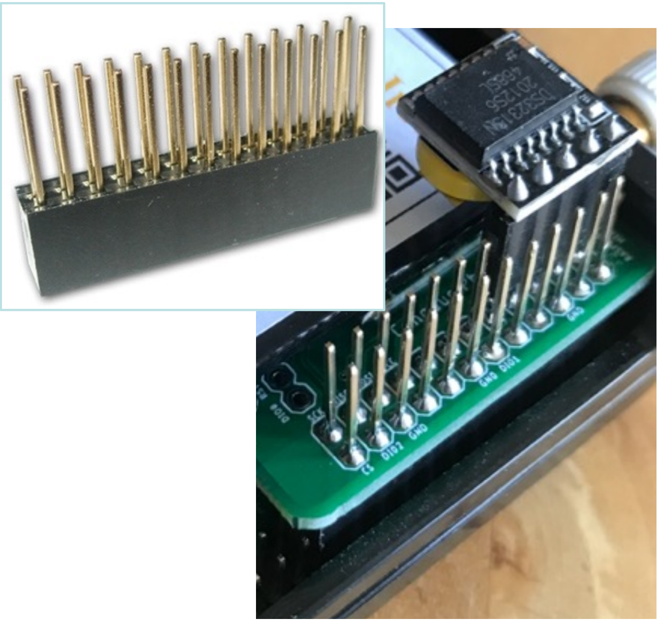
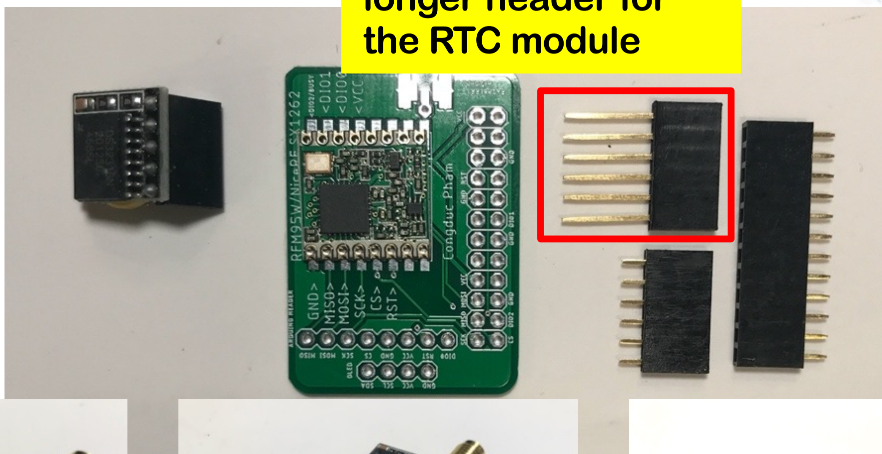
Connecting an RTC module

- Several possibilities depending on availability of components

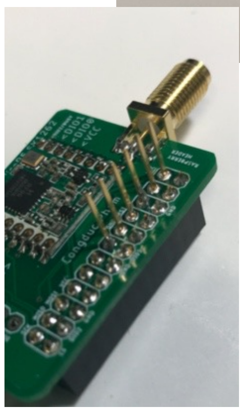


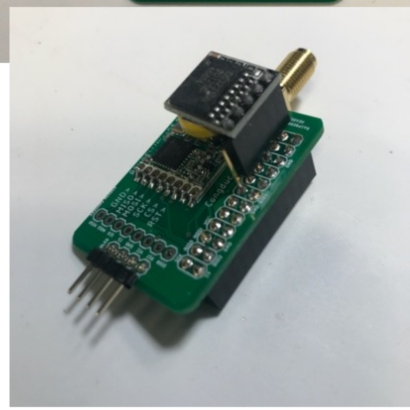


1/ Solder a header with longer pins on the radio hat. You can then cut the remaining pins if you want



2/ use only a 6-pin longer header for the RTC module





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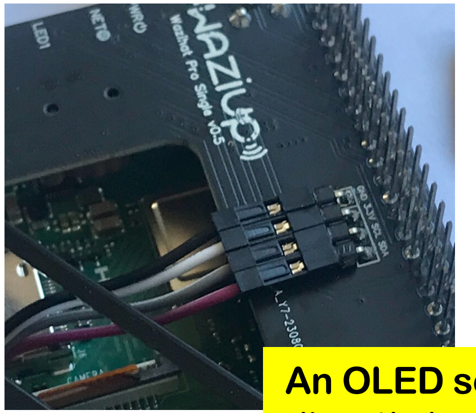
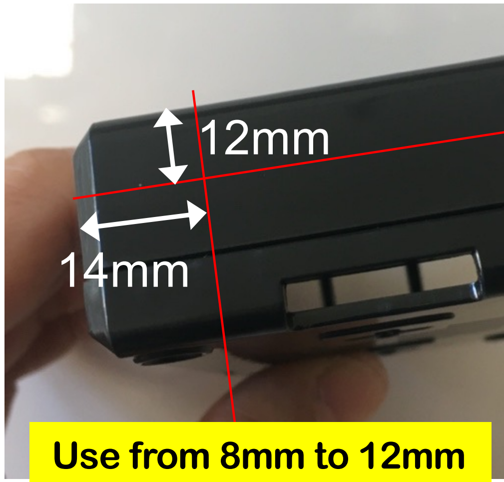
Using WAZIUP LoRa gateway hat

LoRa gateway hat
with on-board RTC,
no need for external
RTC module



An OLED screen can
directly be connected
to the radio hat



Use from 8mm to 12mm
drill bit for metal

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Multi-channel/Multi-SF

- It is possible to have a muti-channel/multi-SF gateway by connecting a LoRa radio concentrator such as the RAK2245 or the new RAK5146 with the RAK5146 Pi Hat (right photo)
- The LoRa radio concentrator will be automatically detected
- IMPORTANT: a reliable 5V 2.5A power source is necessary

RAK2245

RAK5146

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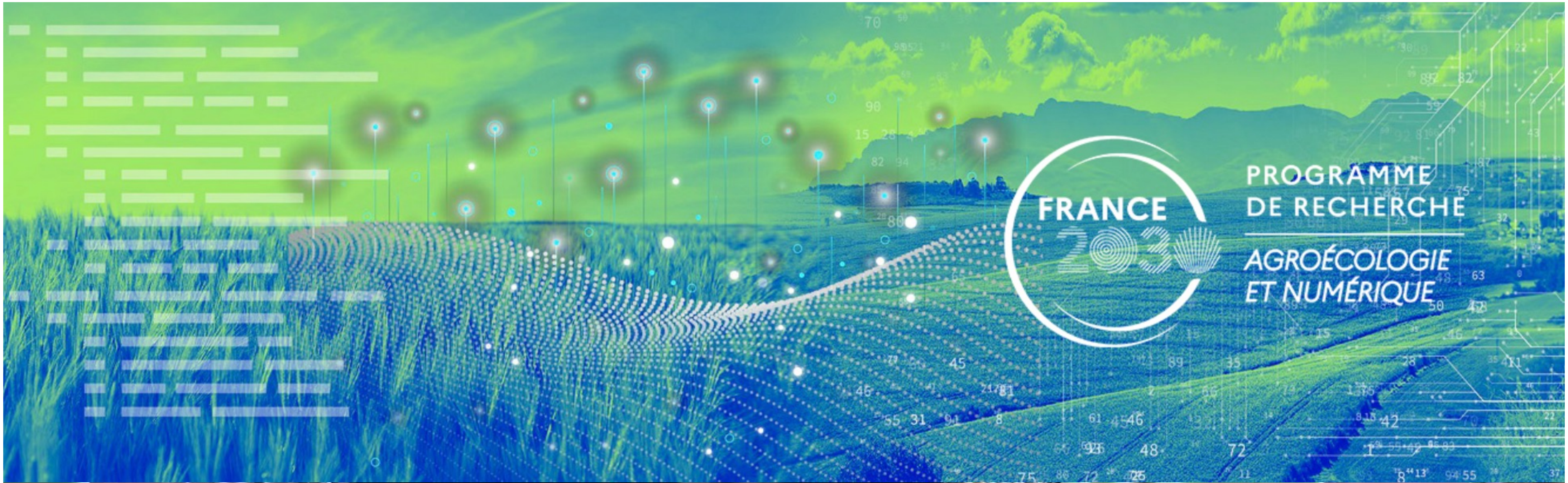

RUNNING THE GATEWAY

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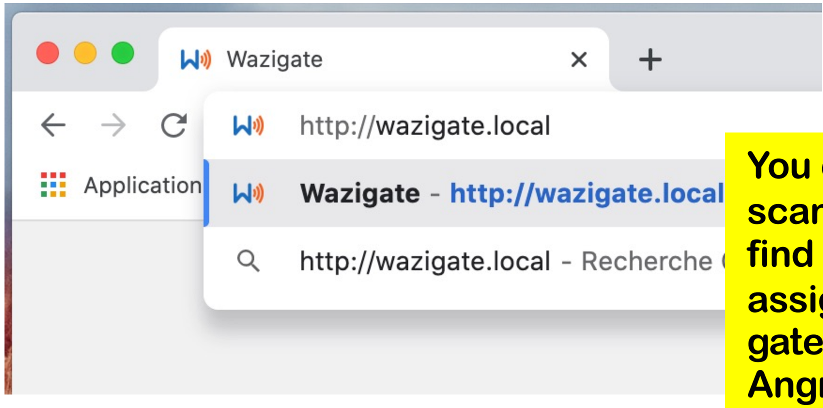
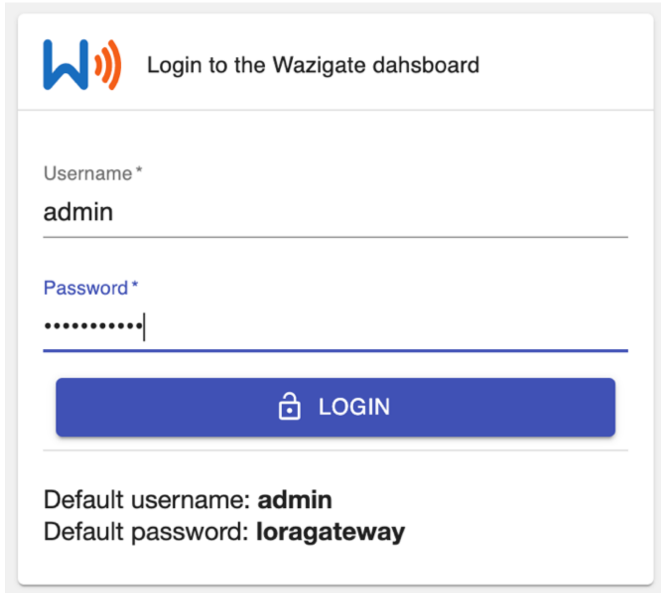
Checking the gateway (Ethernet cable)

- Connect the gateway to your laptop which has Internet
- Enable Internet sharing, laptop provides IP address to gateway
- Power the gateway, wait 3-4mins for boot process
- Open web navigator. Go to http://wazigate.local or use IP address



You can use an IP scanner program to find the IP address assigned to your gateway (such as Angry IP Scanner)



- Use default login to connect
  - User: admin
  - Password: loragateway

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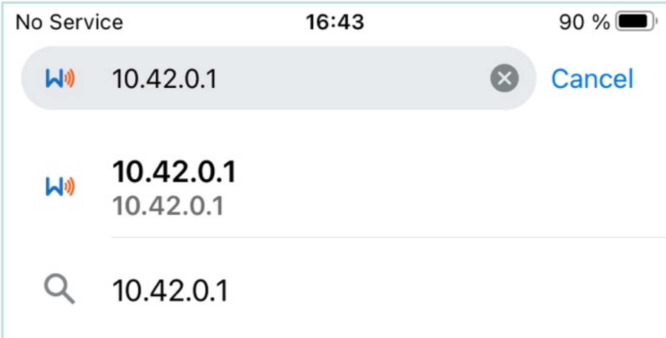
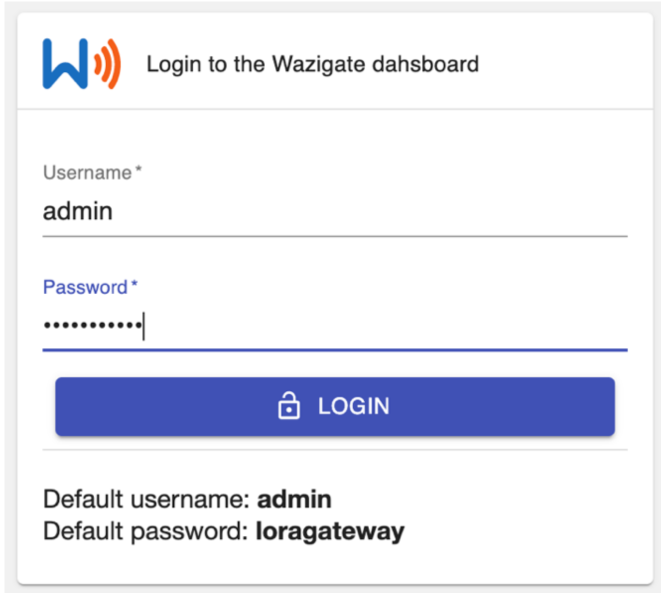
Checking the gateway (WiFi)

- Use a smartphone to check access to gateway through WiFi
- Connect to WAZIGATE_XXXXXXXXXXX WiFi network
  - default WiFi password is loragateway
- Open web navigator. Go to http://wazigate.local or http://10.42.0.1





- Use default login to connect
  - User: admin
  - Password: loragateway

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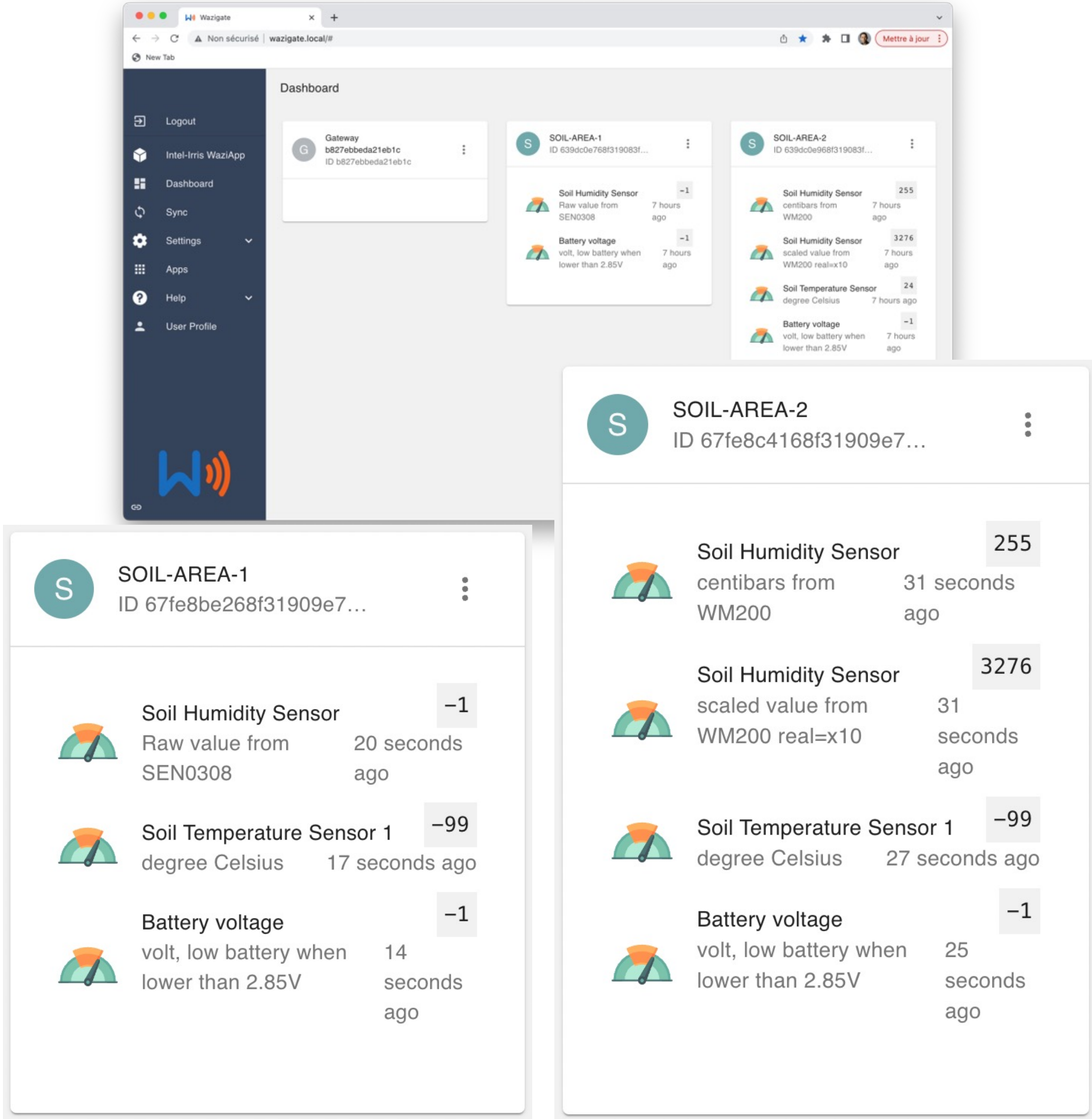
Default gateway configuration (1)

- The default configuration will be ready for
  - 1 capacitive sensor named SOIL-AREA-1 with address 26011DAA
  - 1 tensiometer sensor named SOIL-AREA-2 with address 26011DB1
- Capacitive device will show humidity, soil temperature (if any) and battery values
- Tensiometer device will show centibar, raw resistance, soil temperature and battery values



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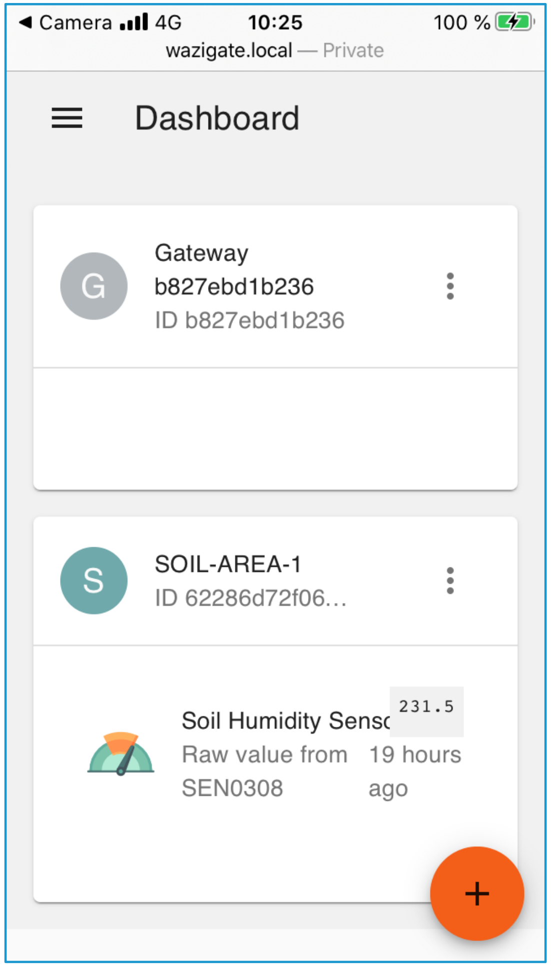
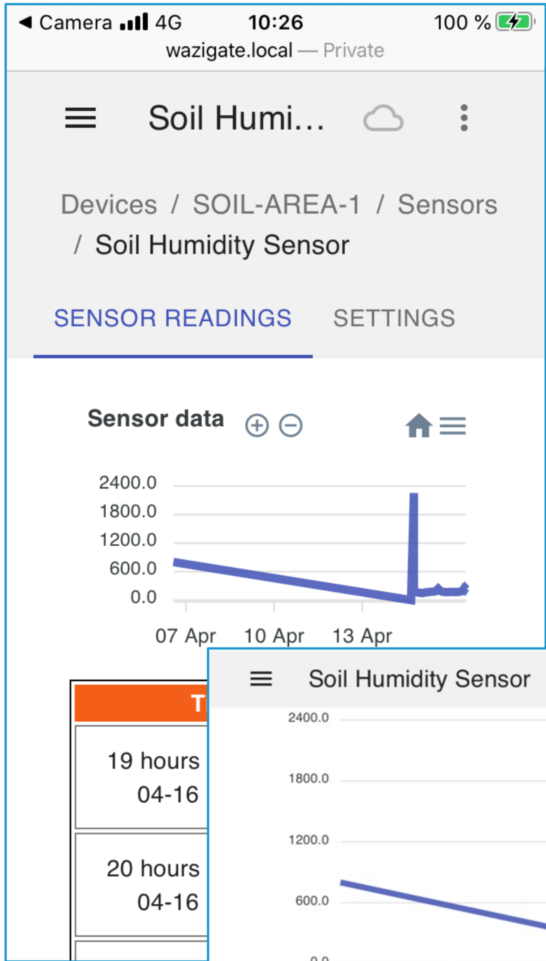
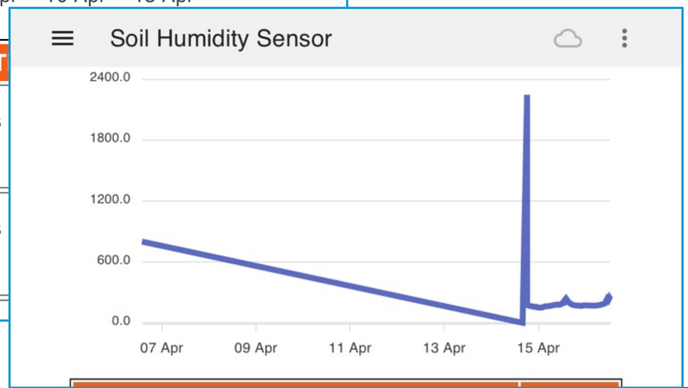
All these steps with a smartphone







http://wazigate.local

http://10.42.0.1

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QR code for connecting to WiFi

- The gateway WiFi is WAZIGATE_XXXXXXXXXXXX where XXXXXXXXXXXX will be the MAC address of the RPI
- For instance WAZIGATE_B827EBD1B236

- With the OLED, a QR code for joining the WiFi network is dynamically generated at boot time and displayed for 10s before the main screen so that users can automatically join with a smartphone
- Then, users can scan static QR code to connect to the gateway's dashboard

http://wazigate.local

http://10.42.0.1

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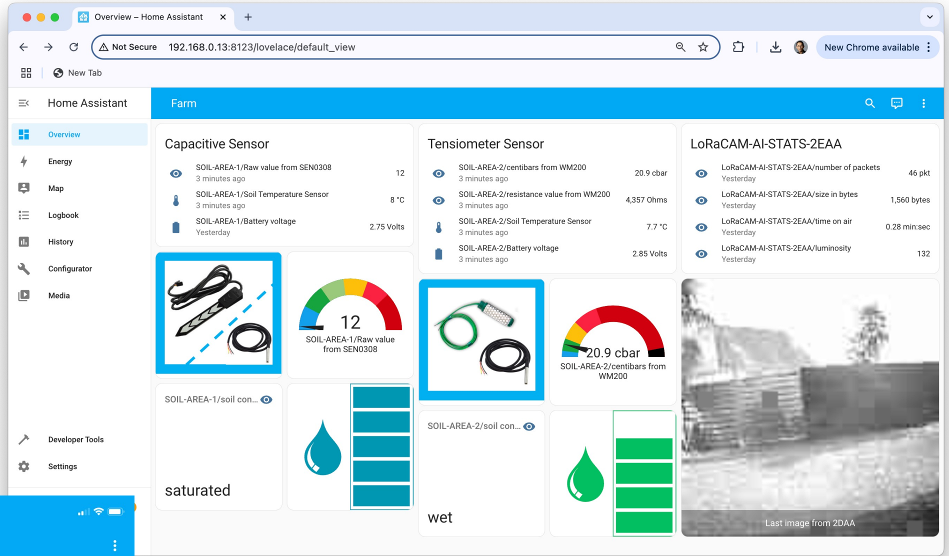
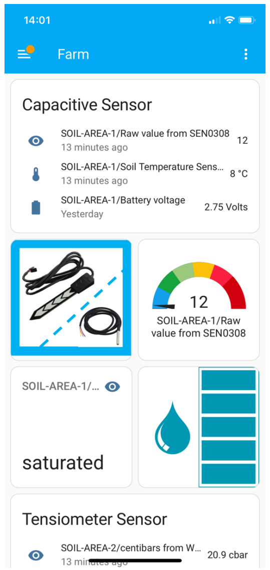
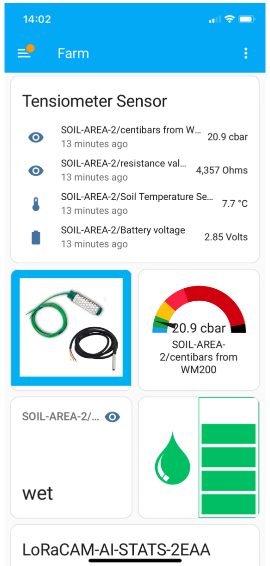
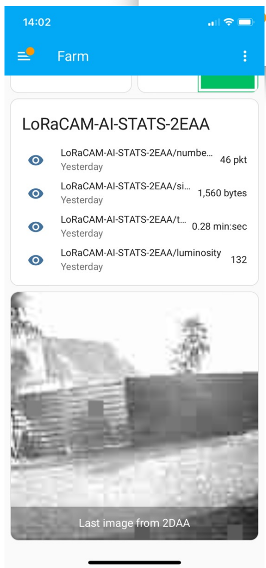
Home Assistant dashboard

- Connect to
  - http://wazigate.local:8123 (Ethernet)
  - http://10.42.0.1:8123 (gateway's WiFi)
- The HA login page should appear
  - User: intelirris
  - Password: intelirris









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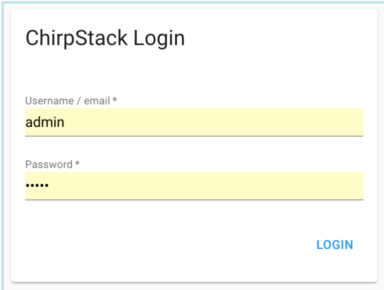
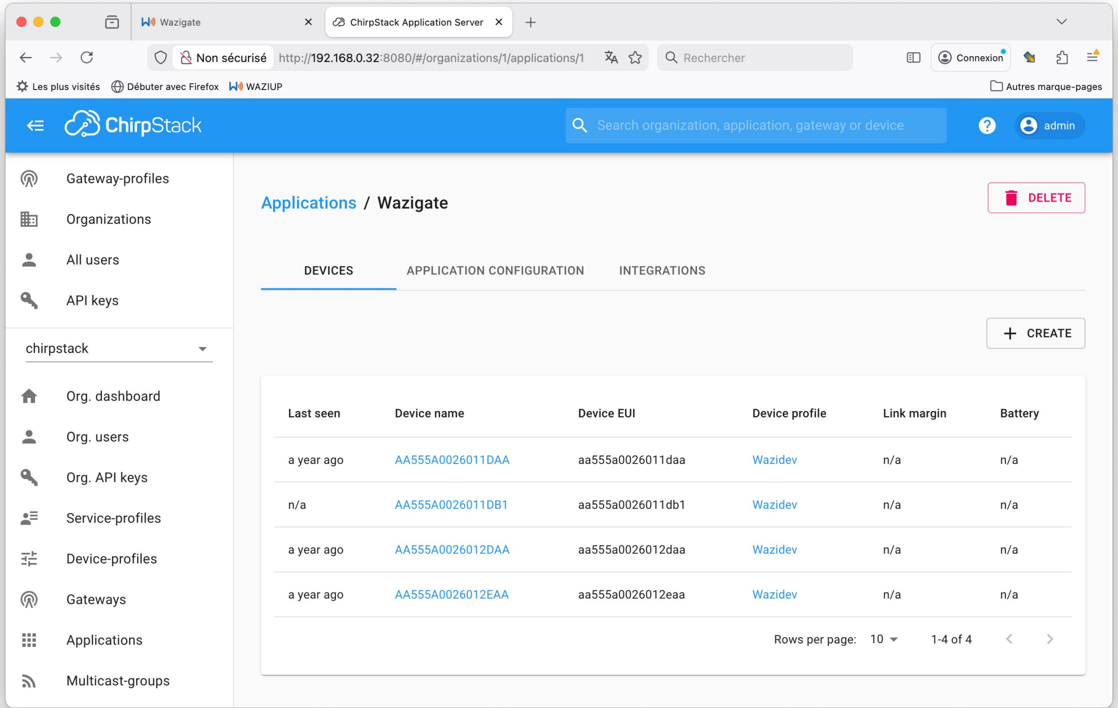
Connect to ChirpStack Network Server

- The generic WaziGate comes with ChirpStack Network Server
- Open web navigator. Go to http://wazigate.local:8080 or use IP address



- Use default login to connect
  - User: admin
  - Password: admin
- Then get access to
  - registered devices
  - live LoRaWAN packet capture
  - downlink, …



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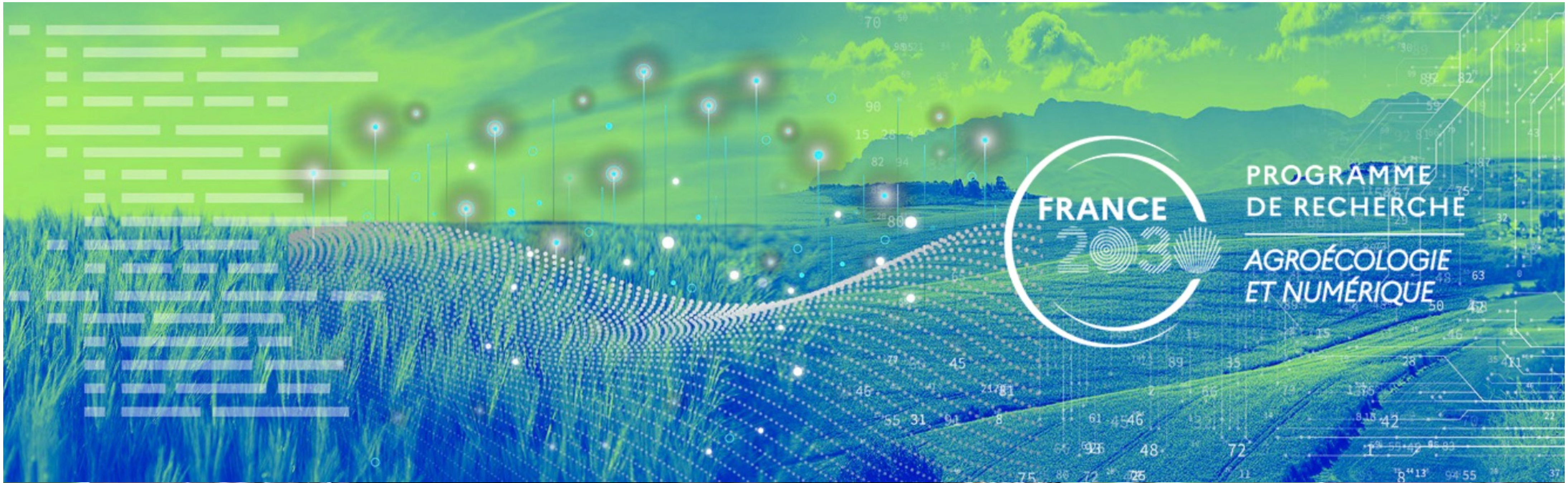

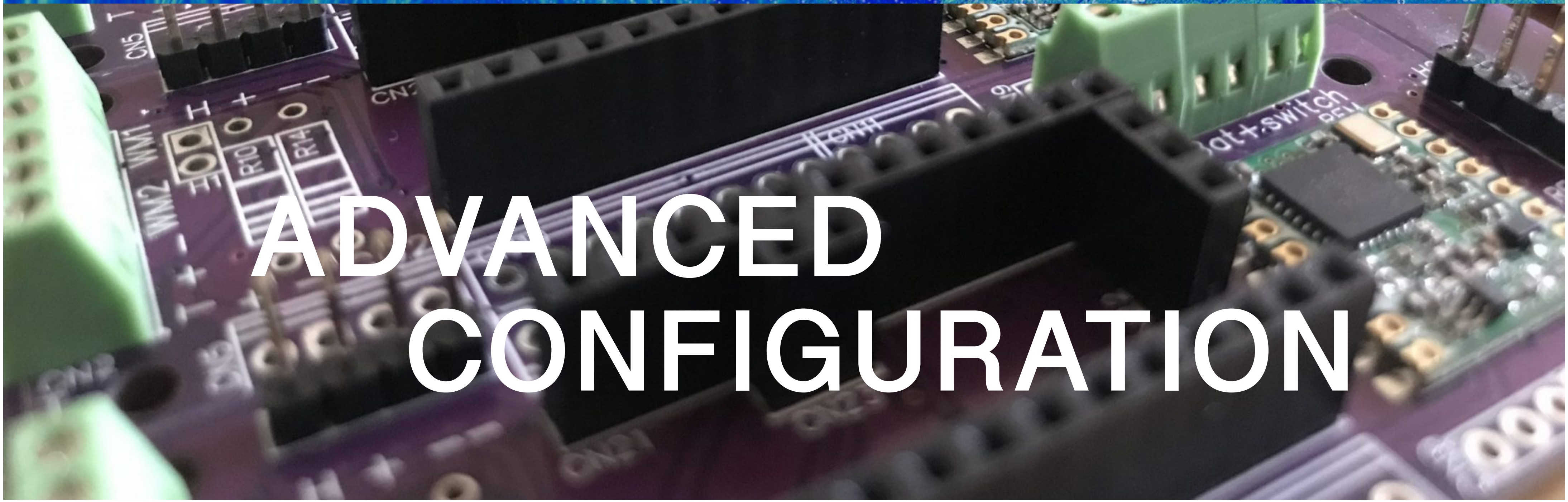

ADVANCED CONFIGURATION

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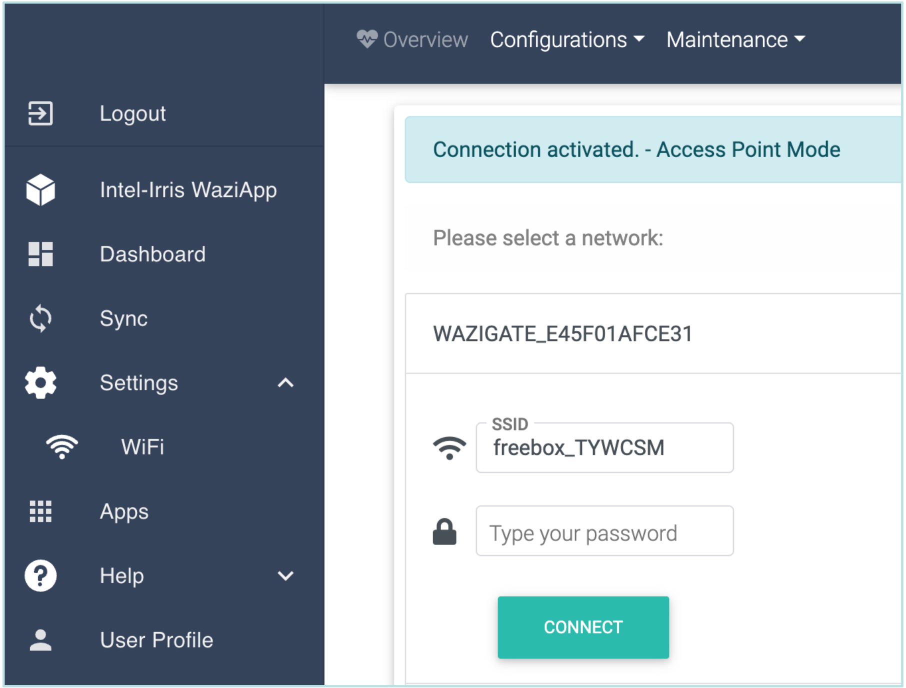
Advanced configuration
connect gateway to a WiFi network

- By default, the gateway acts as a WiFi Access Point
- To connect the gateway to a WiFi network, go to Setting/WiFi to list all available WiFi networks
- Then select the one you want in order to provide the WiFi password
- Ex: connect to freebox_TYWCSM
- Once connected, gateway is in WiFi Client mode



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Advanced configuration

connect gateway to a WiFi network, con't

- You can connect to several WiFi networks, one after another, to have a list of known WiFi networks
- They will be memorized and if the current WiFi network is not available, another available network in the list of known WiFi networks will be selected
- If there are no available WiFi networks in the list of known WiFi networks anymore, then the gateway switches back to Access Point mode

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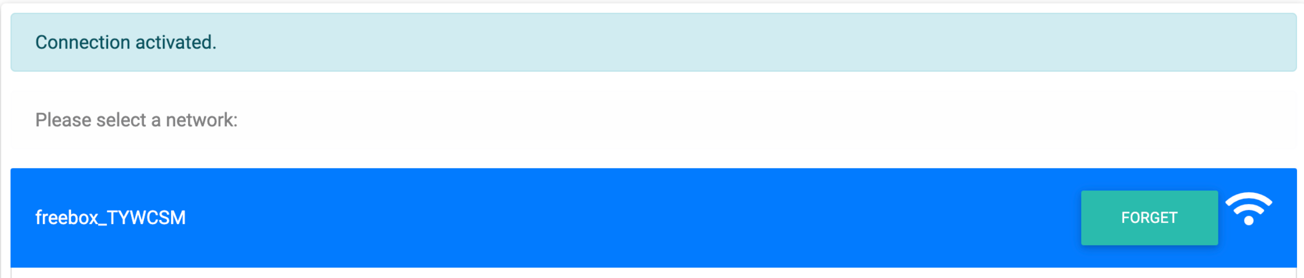
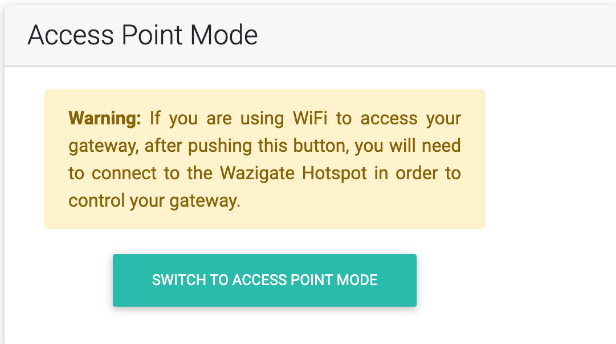
Advanced configuration
switch gateway back to WiFi access point mode

- To get back to Access Point mode, got to Setting/WiFi and simply click on "Forget" for the current WiFi network



- If you previously added several WiFi networks, click on "Forget" for ALL known & memorized WiFi networks
- DO NOT USE the "SWITCH TO ACCESS POINT MODE" option
- IT IS NOT STABLE!



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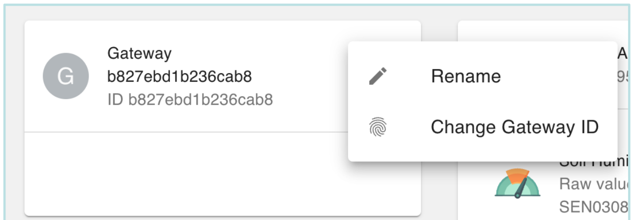
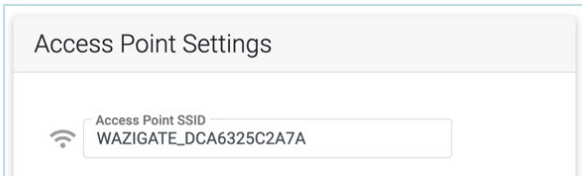
Advanced configuration

Synching your gateway to the cloud

- If you want to sync your gateway to the WAZIUP Cloud (WaziCloud), look at this tutorial page
  - https://www.waziup.io/documentation/wazigate/v2/install/#registration-with-the-cloud
- You will need an account on WaziCloud
  - If you don't have one, you need to create one first
  - https://dashboard.waziup.io/
- Then, you NEED to change your gateway id
  - Use the unique MAC address of your gateway that appears in Settings/Configuration (it is used as your gateway's WiFi hotspot)
  - Here: DCA6325C2A7A
  - Add 0000 at the end to have 16 digits
  - -> DCA6325C2A7A0000

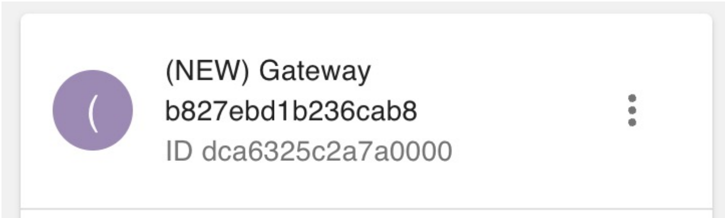


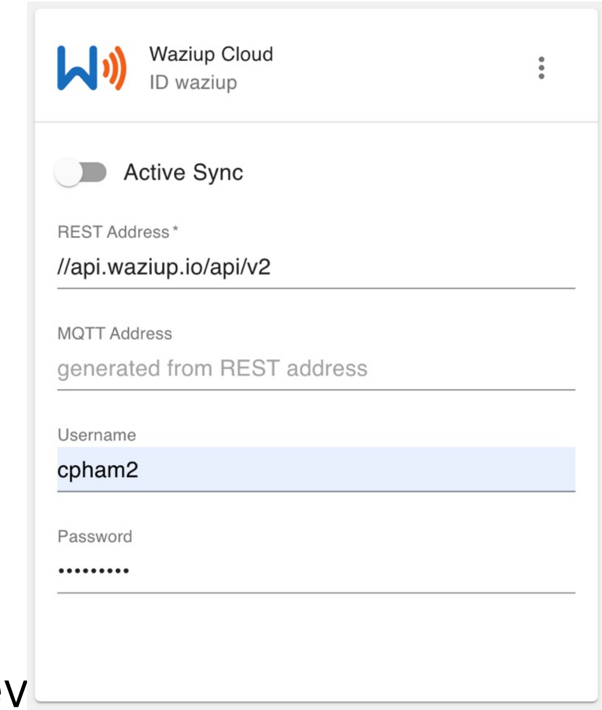


Advanced configuration

Synching your gateway to the cloud, con't

- You should have a new gateway on your dashboard with the new ID





- Enter your WaziCloud credentials in the Sync menu
- Then, just activate sync on your gateway which needs to be connected to Internet
- Log in the WaziCloud dashboard and check that you see your gateway and your dev
- You can activate/deactivate synchronization at anytime

Advanced configuration

MQTT integration from WaziCloud

- With sensor data on WaziCloud, it is possible to subscribe to those data with MQTT protocol
- With command line `mosquitto_sub`
  - moquitto_sub
    -L "mqtt://api.waziup.io/devices/<deviceID>/sensors/<sensorID>/value"
  - mosquitto_sub
    -h api.waziup.io -t devices/<deviceID>/sensors/<sensorID>/value
- With other MQTT integration client/platform
  - Host: api.waziup.io
  - Topic: devices/<deviceID>/sensors/<sensorID>/value
- Output
  - `{"value":34,"timestamp":"2022-12-23T10:23:54Z"}`

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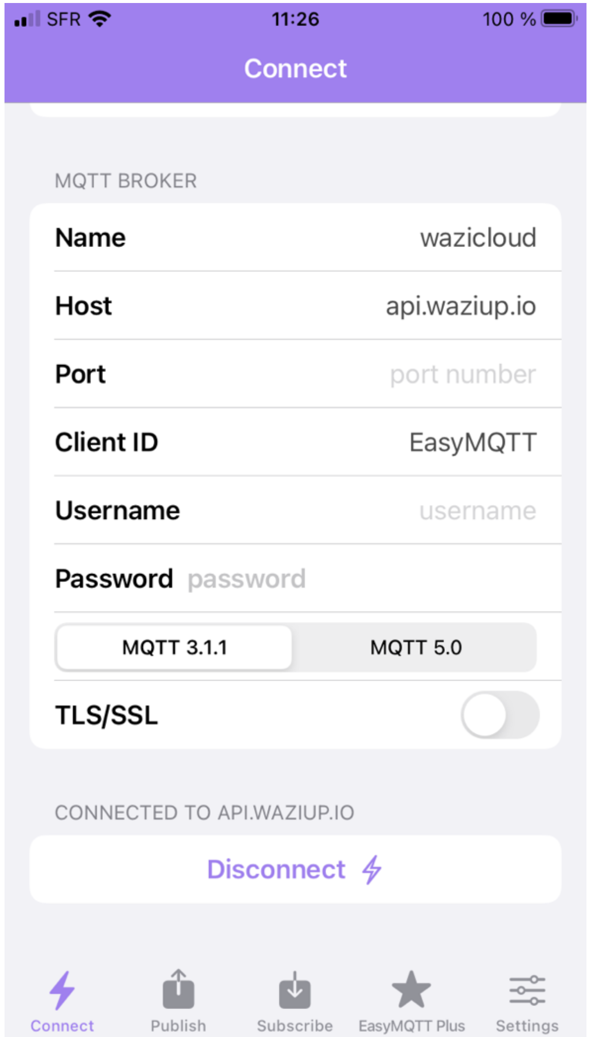
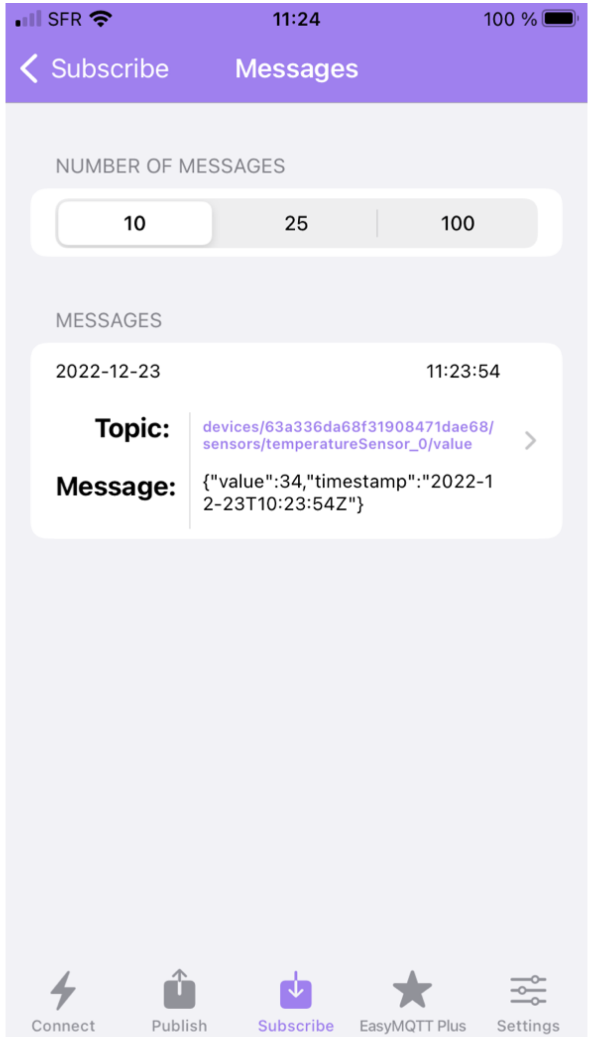
Advanced configuration

MQTT integration – MQTT client on a smartphone

- Example with an MQTT client (EasyMQTT) on an IPhone7





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